AtkinsRéalis

Jiayi Ding, PhD, PE, PMP

Lead Structural Engineer, Technical Manager



Total years of experience

12

Years with firm

3

Education

Ph.D., Structural Engineering, Florida International University, 2026

M.S., Structural Engineering, University of Illinois at Urbana Champaign, 2016

B.S, Civil and Structural Engineering, Michigan Tech, 2014

Registrations/licenses

Professional Engineer:
California: 90653 (since 2019)
Colorado: 64996 (since 2023)
Florida: 88211 (since 2019)
New York: 105285 (since 2021)

Certifications

Project Management Professional (PMP): 2801313 (since 2020)

Professional affiliations

American Society of Civil Engineers (ASCE)
Structural Engineering Institute (SEI)

- Bridge & Tunnel Maintenance, Inspection, & Rehabilitation Committee

Jiayi is a senior structural engineer with 12 years of experience in design, analysis, assessment and rehabilitation of bridges. His technical expertise lies in precast prestressed concrete, post-tensioned spliced girder, segmental, steel plate and tub girders, bridge substructures, and foundations. He is proficient in finite element analysis, dynamic analysis, vessel collision analysis and soil-structure-interaction. He also has substantial experience developing bridge maintenance strategies and asset management, emergency inspection and repair efforts following severe damage or extreme events. He is a recognized contributor to the profession, having presented at national conferences and published in structural engineering journals. He manages a group of 10 structural engineers at AtkinsRéalis in Florida.

Jiayi Ding's project experience includes:

Colorado Department of Transportation (CDOT) Region 1, I-70 (Floyd Hill to Veterans Memorial) Tunnels Design Engineering Services, Final Design & CDMG Services, Clear Creek County, CO. *Engineer of Record.* This project improves capacity and geometry along westbound I-70 and includes six bridges, notably Bridge A: a cast-in-place, post-tensioned segmental structure built by balanced cantilever with 280-ft and 347-ft spans. Jiayi led the IDC for the segmental portion, verifying superstructure and substructure designs, including twin-leaf piers and PT hammerhead piers. He also performed moment-curvature analysis, checked secondary effects, and evaluated cracked-section behavior and reinforcement stresses. Duration: 10/2024-5/2025

Central Florida Expressway Authority (CFX), SR 414 Expressway Extension (US 441 to SR 434) Project #414-208, Orlando, FL. *Lead Structural Engineer*. AtkinsRéalis is Lead Designer (nearing 90%). Four bridges are included in the project, but the design effort centers around the 58-span elevated 4-lane viaduct (8638' total length) with 89' bridge width. Six spans will have four curved steel box girders with variable depth webs, varying 108" to 72" deep (from 9' at midspan to 6' at the supports) and spans up to 300' long. This is believed to be the first curved, steel box with variable depth webs designed in Florida. The remaining units are typical spans of six 78" deep prestressed concrete U-beams with spans up to 160'. The project is estimated to cost $500 million and the design schedule is 30 months. The proposed inverted-T pier design lowers the profile by 10' through the residential community and represents a $60 million cost savings when compared to the PD&E segmental bridge concept. Jiayi led the substructure and foundation design of the 2 end bents and 56 piers, overseeing the design development and plan production of hammerhead and inverted T piers. Due to the unique geometry and loading carrying requirements, the two types of bridge substructure were designed and checked based on strut and tie method and refined finite element method in addition to conventional reinforced concrete design. He also led the design of deep foundation system utilizing steel H-piles and pipe piles that can withstand sinkhole extreme event loading and soil behavior. Duration: 11/2023-1/2026

Georgia Department of Transportation (GDOT), SR 4 BU/US 1 BU/ US 23 BU at CSX Railroad Ware County, GA. *Quality Control Reviewer*. This project aims to construct a grade-separated bridge to eliminate delays caused by trains blocking the existing at-grade railroad crossings. The new bridge (15 spans, 1700 ft total length)

will reduce travel times and minimize the risk of train-vehicle and train-pedestrian interactions along Ossie Davis Parkway. Additionally, it will provide a grade-separated crossing over an associated spur line. Several intermediate bents of this bridge are post-tensioned inverted T straddle bents, which requires refined modeling and analysis. Jiayi performed Quality Control (QC) review for the design of post-tensioned straddle inverted-T bents, ensuring accuracy and compliance with project standards. His responsibilities included verifying structural models in MIDAS and reviewing calculations related to service stress checks, flexural moment capacity, shear and torsion demands, and corbel design. Jiayi checked design against construction plans to ensure consistency and constructability. Duration: 9/2024-9/2025

Fort Lauderdale-Hollywood International Airport Terminal 5, Broward County, FL. *Structural Engineer*. The Terminal 5 expansion at Fort Lauderdale-Hollywood International Airport includes a new flyover bridge (4-span continuous steel plate girders, incorporating two 99 ft side spans, two 165 ft interior spans, with a total length spanning 528 ft) carrying traffic from the new concourse to Terminal Drive. With a tight 160-ft radius and shallow girder depth, the structure required refined analysis and specialized detailing to meet geometric and clearance constraints. Jiayi served as the lead designer, advancing the bridge from conceptual through final design. He prepared the Bridge Development Report, evaluated steel I-girder and box girder alternatives, and designed the hammerhead piers, end bents, and deep foundations using 24-inch driven piles. He also performed high-fidelity finite-element modeling to capture torsion, cross-frame behavior, and uplift forces, designing reinforced end diaphragms to ensure stability and serviceability. Duration: 9/2022-9/2023.

Ontario Line TA Planning, Metrolink, Thorncliffe Segment, Toronto, Canada. *Segmental Bridge Designer*. The Don Valley Crossing is five-span balanced-cantilever segmental bridge spanning a total length of 505 meters (1657 ft) with the main span measuring 138 meters (453 ft). The reinforced concrete substructure consists of three twin-wall piers and two single-column piers to mitigate the creep and shrinkage effects stemming from an unbalanced span layout and various construction stages, the bridge design involved refined analysis based on substructure's cracked section stiffness. This type of refined analysis considering the flexible twin-wall piers in accommodating creep, shrinkage and negative unform temperature is not addressed in AASHTO publications. Jiayi led a structural design team through a series of advanced analyses to support the design and evaluation of segmental bridge substructures. The unique twin-wall piers requires high fidelity moment curvature analysis to determine the effective stiffness based on cracked section analysis which is not commonly addressed in AASHTO code. His responsibilities included performing load calculations and assessing moment amplification effects on substructures due to secondary load paths. He conducted cracked section analysis of columns and evaluated tensile stresses in column reinforcement during both construction and service stages. Utilizing Xtract software, Jiayi performed detailed nonlinear section analysis to ensure accurate modeling of structural behavior and reinforcement performance under realistic loading conditions. His work contributed to the development of robust and reliable substructure designs that meet project requirements and industry standards. Duration: 9/2022-9/2023

Florida Department of Transportation (FDOT) District 4, Roosevelt Bridge Emergency Repair, Stuart, FL. *Independent Design Reviewer*. During a biennial inspection, FDOT identified significant cracking on the southbound Roosevelt Bridge and assembled a statewide structural team to assess the issue. Jiayi conducted an independent design review of the proposed retrofit for the damaged unit (consists of 5 balanced cantilever continuous spans, with span lengths vary from 196 ft to 260 ft), analyzing stresses, moment capacity, shear, and torsion to validate the effectiveness

of new external post-tensioning. He also reviewed tendon deviator details, constructability, bridge jacking procedures. Duration: 6/2020-10/2020.

Florida Department of Transportation (FDOT) District 4, SR A1A North Bridge over ICWW Design Services, St. Lucie County, FL. *Substructure Lead Designer*. This project replaces the existing bascule bridge with a new high-level fixed bridge over the ICWW, providing 85-ft vertical and 125-ft horizontal clearance at an estimated construction cost of $111.5 million. The 4,152.5-ft bridge includes 23 simple spans and a three-span continuous channel unit with Florida I-beams. Jiayi led substructure design from 60% to final, performing vessel-collision and scour analyses, designing water-pier substructures and foundations, and coordinating with structural and geotechnical teams throughout the project duration. Duration: 12/2019-12/2021.

Florida Department of Transportation (FDOT) District 4, SR 710/Beeline Highway (Northlake Boulevard to Blue Heron Boulevard) Design Services, Palm Beach County, FL. *Structural Engineer*. This project widens SR 710 from a four-lane rural highway to a six-lane urban facility and includes replacing the SR 91 bridge over SR 710 with a phased-construction, five-span FIB beam bridge (79–118 ft spans). Jiayi designed the multi-column-cap piers, reviewed PC/PS I-girder designs, and performed construction-phase load rating using LEAP Bridge. He also designed mile-long CIP cantilever retaining walls supporting the junction slab and sidewalk using FDOT Mathcad. Duration: 12/2018-4/2022.

Broward County Department of Public Works, General Engineering Consultant, Broward County, FL. *In-house consultant and Project Manager*. After the partial collapse of an FDOT Sonovoid bridge in 2017, Broward County launched a rehabilitation program for more than 100 similar structures. Jiayi led the effort as an in-house consultant, developing a near-surface-mounted-dowel repair method and validating its effectiveness through load testing. He directed structural evaluations, prioritized repairs, performed structural design, cost-estimate and coordinated with multiple agencies throughout design and construction phases. Duration: 5/2021-10/2022.

Florida Department of Transportation (FDOT) District 5, I-75 Phased Design Build Auxiliary Lanes from SR 44 to SR 200, Sumter and Marion Counties, FL. *Quality Control Reviewer*. The project involves four bridge replacements over I-75 (CR 462, CR 475 and SW 66th Ave) and the widening of the I-75 bridge over SR 44. AtkinsRéalis provided engineering services for structures, roadway, drainage, environmental/permitting, sign and pavement markings, ITS, utilities, landscaping and aesthetics, and other roadway features. Jiayi performed Quality Control (QC) review of the structural design calculations and plans of 2 bridge replacement (CR 462 over I-75, and SW 66th Ave over I-75). Duration: 11/2025-5/2026.

Florida Department of Transportation (FDOT) District 5, I-4 Ultimate Design-Build, Orlando, FL. *Structural Designer.* FDOT District 5's $2.3B I-4 Ultimate design-build project reconstructs 21 miles of I-4 through Orlando, including 15 interchanges, 145 bridges, and 85 new lane-miles. Jiayi provided post-design support for four complex steel I-girder and steel box-girder bridges, reviewing shop drawings, performing erection and load-rating analyses, and conducting advanced LARSA 4D modeling to evaluate stresses near girder openings. His work guided strengthening at bearing soleplates, deck pour sequencing, and load-rating updates, while preparing plan revisions and responding to RFIs/RFMs. Duration: 6/2018–12/2020.

Florida Department of Transportation (FDOT) District 6, I-395 Signature Bridge, Miami-Dade County, FL. *Structures Engineer*. This large-scale design-build project

includes the construction of a high-profile signature bridge across the Biscayne Boulevard. The signature bridge will have the largest precast concrete segmental arch in North America. The bridge's tallest arch will soar 325 feet above the ground and span 650 feet across. As a structural engineer on the Signature Bridge design team, Jiayi Ding was responsible for the detailed analysis and design of key arch components and supporting structures. His work included longitudinal design of Arches 1 and 4, addressing normal stress, principal stress, ultimate flexural capacity, shear, and combined shear-torsion effects. He also performed transverse analysis of precast post-tensioned arch segments under cable-induced loads to ensure structural performance and stability. Jiayi completed the analysis and design of cast-in-place reinforced concrete starter segments and designed the end bents, contributing to the overall integrity and constructability of the bridge system. Duration: 1/2019-12/2019

Florida Department of Transportation (FDOT) District 6, SR 25/Okeechobee Road (East of 116th Way to East of NW 87th Avenue), Miami-Dade County, FL. *Structural Engineer*. This project reconstructs SR 25/Okeechobee Road to reduce congestion between NW 116th Way and NW 87th Avenue, including replacement of the NW 121 Way bridge (estimate construction cost is $105.6 million). The bridge work involved demolishing the existing deck and constructing a widened, highly curved reinforced-concrete flat slab on the existing substructure. Jiayi developed a refined plate-element model to analyze curved-slab behavior, capturing moment and shear demands that exceeded AASHTO strip-method results. His analysis improved load-path accuracy considering skew, supporting the slab's reinforcement design. He also verified the existing substructure for new load demands. Duration: 9/2019-6/2021.

Florida Department of Transportation (FDOT) District 6, Palmetto Expressway Express Lanes, Miami-Dade County, FL. *Structural Engineer*. The SR 826/Palmetto Expressway Capacity project widens the corridor to reduce congestion and includes modifying the flyover ramp bridge to accommodate a new general-purpose lane. The work involves partial deck removal, exterior girder extraction, hammerhead pier demolition, and reconstruction of the existing C-pier. Jiayi led feasibility study, performing girder analysis with MDX and developing a global construction-stage model to evaluate load transfer between existing and new C-piers. He designed deck reinforcement, checked pier flexure and shear, and prepared conceptual C-pier and foundation layouts considering redistributed loads. Duration: 8/2020-12/2020.

New York City Department of Transportation (NYDOT), Queensboro Bridge Upper Roadway Deck Replacement, Queens, NY. *Structural Engineer*. The Queensboro bridge opened to traffic on June 18, 1909, as the longest cantilever bridge in the United States. This five-span cantilever truss bridge, designed to accommodate heavier loads, is the only one of the four great East River Bridges that is not a suspension bridge. This project involves providing design and construction support for the reconstruction of the upper deck on main spans. With 200,000 square feet of roadway deck replacement needed due to years of overloading, the challenge lies in selecting a lightweight, durable replacement for extended service life. To address this, a novel open-rib steel orthotropic deck with a polymer overlay was designed, maximizing stiffness-to-weight ratio for the long span cantilever truss bridge. Jiayi was responsible for the structural assessment and retrofit design of an open-rib orthotropic deck system, with a focus on fatigue performance under cyclic loading conditions. His work included evaluating maximum von Mises stresses through detailed fatigue analysis and developing retrofit solutions for critical components such as floor beams, sub-floor beams, support diaphragms, and their connections. Jiayi performed nonlinear finite element fatigue analysis using ANSYS Mechanical to capture complex stress behavior and fatigue life predictions. In addition, he designed an innovative, customized steel traffic railing system that met AASHTO TL-5 crash testing

requirements, ensuring both safety and compliance with national standards. Duration: 6/2016-10/2017

New York Metropolitan Transportation Authority (MTA), Seismic Retrofit of Approaches at Throgs Neck Bridge, Bronx, NY. *Structural Engineer*. Jiayi served as the main investigator for the Throgs Neck Bridge Approach Viaducts Rehabilitation project, focusing on seismic vulnerability and rehabilitation design to extend the viaducts' service life by 25 years. He reviewed past reports, calculated the characteristics parameters of seismic isolation bearings, and conducted comprehensive modeling and analysis, including nonlinear time-history and response spectrum analyses. Jiayi also co-authored the technical report and coordinated project tasks. Duration: 9/2016-10/2017

New York Metropolitan Transportation Authority (MTA), Throgs Neck Bridge Suspended Span Deck Replacement, Bronx, NY. *Structural Engineer*. The scope of this project was to design the replacement of the deck system on the suspended spans of the Throgs Neck Bridge. The existing deck system, a concrete-filled grid with relief joints at every sixth-floor truss, was approaching the end of its useful service life. In addition, the relief joints are leaking, causing corrosion on the stringers, floor trusses, and lateral bracing below. Thus, a new closed-rib orthotropic deck system was designed. A preliminary wind study and analysis were conducted to check the bridge stability against a 10,000-year return period. The predicted mean hourly wind speeds at the deck level of the two joint-less deck alternatives far exceeded the 10,000-year wind speed (115 mph), meaning the bridge is very stable against strong winds. Wind tunnel testing will also be performed for the proposed structure, which will include the impacts of staging and wind on trucks. Jiayi was responsible for the structural analysis and design of a steel closed-rib orthotropic deck system subjected to fatigue truck loading. His work included evaluating fatigue performance and stress behavior under repetitive loading conditions, ensuring long-term durability and compliance with design standards. He also designed strengthening measures for floor beams and their connections to enhance structural capacity and service life. Additionally, Jiayi performed nonlinear analysis and developed a customized steel traffic railing system that meets AASHTO TL-5 crash testing requirements, combining safety, innovation, and structural efficiency. Duration: 6/2016-10/2017